Delto Gripper

DG – 5F

User Manual

English

Ver. 1.1.4



INFORMATION

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Notice of revisions to the user guide

This documentation is subject to revision for technical improvements.

Manual Version: 1.1.4 English (2025.06)

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Chapter 1. General Information

1.1 User Manual Information

- This document provides information for the safe and proper use of DG-5F.
- This document must be accessible to the user at all times as it contains integrated information, including safety guidelines for safe operation of DG-5F.
- Images are used in this document to help the user understand the design of parts that are different from the actual product and complicated parts.
- This also applies to the guidelines contained in the documents listed in Section '1.1.2 Documents provided by TESOLLO'

1.1.1 Danger and warning signs

⚠ WARNING: Risk of Electric Shock

- Contact with live electrical parts may result in death.
- Turn off the power supply before performing assembly, adjustment, or maintenance.
- Ensure power is completely disconnected and no residual current remains before starting work.
- Only authorized personnel may enter or operate within the work area.

⚠ CAUTION: Risk of Entanglement

- Keep body parts clear of moving components during operation.
- Serious injury may occur if clothing or limbs become caught.

⚠ CAUTION: Hot Surface

- Surface may remain hot during and after operation.
- Avoid contact to prevent burns.

⚠ WARNING: Risk of Falling Objects

- Serious injury or death may result from falling objects held by the equipment.
- Do not exceed the maximum grasping load.
- Do not enter the equipment's operating area during operation.
- Never leave grasped objects unattended.
- Always wear appropriate personal protective equipment (PPE).

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1.1.2 Documents provided by TESOLLO

- TESOLLO product catalog data sheet
- 3D modeling & 2D drawing
- User Manual for the product
- Other product-related information

※ These documents can be downloaded from en.tesollo.com.

1.2 Target Model for this Document

- DG-5F

1.3 Warranty

- One-year warranty is provided from the delivery date of the gripper, if it is used for its intended purposes while complying with the following regulations.
  - Compliance with the operating limits
  - Compliance with operating environment and working conditions

  See '3.1 DG-5F Basic Information'
- Components in contact with target objects and worn-out components are excluded from the warranty.

1.4 Products Provided

- The products provided include the following.
  - Delto Gripper series
  - User manual
  - Package of components

1.5 Additional Devices

- Additional devices for DG-5F operation are as follows, and some of them are available for purchase independently.
  - EtherNET communication cable
  - USB to RS485 converter
  - 24V power supply (supply of 10A or higher is required)

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Chapter 2. Basic Safety Precautions

2.1 Applications for DG-5F

- DG-5F is developed to hold or fix a target object. Its specific applications are as follows.

A. Pick and Place	**D. Assembly**
B. Bin Picking	**E. Fastening**
C. Machine Tending	**F. A service robot that mixes various tasks**

- DG-5F can be used only for the performances listed in the 'Chapter 3 Technical Information'.
- DG-5F is produced for industrial, service, and research purposes.
- Guidelines for proper use of DG-3F-M are included in this document.

2.2 Improper Application of DG-5F

- The use of DG-5F beyond its performance limit is regarded as improper use.
- See '3.1 DG-5F Basic Information'

2.3 Structural Modification

When structural modifications are attempted

- The use of unauthentic replacement parts may cause human casualties or equipment failure during work. Therefore, only genuine and authentic replacement parts approved by TESOLLO must be used.

2.4 Replacement Parts

When unauthentic replacement parts are used

- The use of unauthentic replacement parts may cause human casualties or equipment failure during work. Therefore, only genuine and authentic replacement parts approved by TESOLLO must be used.

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2.5 Operating Environment and Working Conditions

Operating environment and working conditions requirements

- Improper operating environment or working conditions may cause problems with equipment safety, leading to serious injuries to the user or serious defects in the equipment, such as a reduction in the service life.
- Therefore, the suitable environment described in '3.1 DG-5F Basic Information' must be maintained.

2.6 Recommended User Qualification

Unqualified users

- Unqualified users may cause serious injuries to themselves or damage to the equipment. Therefore, the following regulations must be followed.
  - All tasks must be performed by verified users.
  - Users must familiarize themselves with the assembly of the equipment and study the user manual before they use the equipment.
  - They must comply with domestic and general safety regulations.

Recommended users

- Users with extensive experience and knowledge can perform machine or system-related tasks. They should have the ability to perceive and prevent risks in advance based on guidelines and safety regulations.

2.7 Wearing Protective Gear

Wearing personal protective equipment

- Wearing personal protective equipment can protect users by shielding them from harmful environments and preventing harms of the accidents.
  - Be aware of regulations for safety and accident prevention and wear protective gear recommended for the work environment.
  - Wear goggles, safety suit, and helmets when around moving machinery and environment exposed to toxic chemicals.
  - Wear protective gloves while working with sharp objects.
  - Wear heat-resistant gloves while working with hot objects.

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2.8 Cautions for Safe Use of Equipment

Improper use by the user

- Improper adjustment or assembly may degrade the safety of equipment and cause serious injuries and equipment damage depending on the severity. Thus, the listed guidelines must be complied with.
  - Completely block the sources of interferences with work.
  - Use equipment for its original purpose.
  - Comply with safety precautions and assembly regulations.
  - Avoid being exposed to environment with corrosive materials.
  - Comply with regulations for equipment management and maintenance.
  - Comply with regulations for safety, accident prevention, and environmental protection according to the equipment operation environment.

2.9 Cautions During Operation

Cautions during equipment operation

- Improper handling of equipment during operation may damage product safety and cause serious injuries or product damage.
- When the object held by equipment falls or equipment parts are abnormally released, serious injuries or equipment damage can be caused. Therefore, it is necessary to identify and specify dangerous zones in safe ways.
- Entry into the dangerous areas during equipment operation must be prohibited.

2.10 Cautions in Case of Equipment Failure

- After removing the equipment from the work line, report the failure to the department or person in charge.
- Perform the maintenance and repair of the equipment through TESOLLO.
- Do not use the equipment until the maintenance is completed.

2.11 Cautions During Disposal

Disposal

- Improper disposal of components can cause serious injuries to the user or equipment defects depending on the degree of harm to parts or environment.
- Disposal must be performed properly by complying with local regulations or through decomposition and recycling of equipment components.

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2.12 Cautions for Safety at Work

- Follow the recommended safety distance.
- Do not deactivate safety devices.
- Identify dangerous areas and wear protective gear before work.
- Perform equipment installation, modification, maintenance, and calibration in power off mode.
- Do not approach the equipment when the power supply is switched on.
- Do not enter the equipment workspace.
- Ensure that the emergency off (EMO) button is accessible at all times.

2.12.1 Cautions while handling the equipment

Improper handling

- Improper handling causes defects in equipment safety and may lead to serious injuries to the user or component defects.
  - All tasks must be performed by qualified users.
  - The equipment and user must be protected from dangerous work.
  - Relevant regulations to prevent accidents must be followed.
  - Precautions for moving machinery and collision prevention and the use of appropriate transport equipment are required.
  - Install work zone fences and prohibit entry of persons.
  - Do not work in the absence of the manager.
  - Do not leave the object held in the equipment unattended.

2.12.2 Cautions for dangerous operation

Abnormal operation

- Equipment malfunctions may cause serious injuries to the user. Malfunctions mainly occur in the following situations and emergency power off is required.
  - Power supply is terminated, but equipment is reactivated by residual power.
  - The equipment is operated by an improperly connected drive.
  - Operational mistakes, wrong parameters, and software errors occur.

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2.12.3 Cautions against electric shock

When electronic equipment is in operation

- The following items must be followed because parts in contact with electric current may lead to fatal electric shocks in severe cases.
  - Work with electronic equipment must be performed by qualified electrical and mechanical technicians.
  - Electrical wires must be connected or disconnected only in power-off state and safety is secured.
  - Check the equipment cover and protective devices. Do not touch active parts.
  - Do not touch equipment terminals in power-on state.

Cautions against static electricity

- Equipment parts or connected elements may cause static electricity. Touching static charges may cause a shock and injury.
  - The equipment manager must check whether these elements comply with safety regulations.
  - They must be tested by an electrical technician in the same environment as the actual work conditions.
  - Safety must be ensured through regular inspections.

2.12.4 Cautions for magnetic and electromagnetic fields

During work in magnetic and electromagnetic fields

- Work in magnetic and electromagnetic fields may cause serious injuries.
  - Persons with metal implants or hearing aids should not enter the zones without the consent of a doctor.
  - Do not use high-frequency or wireless devices near the zones.
  - Control systems and drive systems are operated at different intervals.
  - In the presence of high risk in the work zone, the EMC test for environmental verification is required.

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Chapter 3. Technical Information

3.1 DG-5F Basic Information

No.	Contents	DG-5F
Mechanical Spec.		
1	Pinching Payload (Rated, Max) [kg] * Performance based on the use of the thumb and index finger. * It may vary depending on the object's shape or surface material.	2.5, 5
2	Envelop Payload (Rated, Max.) [kg]	10, 20
3	Total DoF (DoF / Finger)	20 (4)
4	Product weight [g]	1,763
5	Maximum force at the fingertip of each finger [N] * 3.1.2 DG-5F Zero Position ref.	97
6	Maximum Graspable Object Size [mm] * The size may vary depending on the weight.	170
7	Default Fingertip Color and Silicone Hardness * May vary depending on the fingertip type.	White + Gray, 40 Shore A
Electrical Spec.		
1	Motor Type	BLDC Motor
2	Rated Voltage [V]	24
3	Maximum Power Consumption [A]	10
4	No-Load Speed of Each Joint [RPM]	75
5	Peak Torque of Each Joint (Stall torque) [Nm] *Developer Mode(1) : Stall torque is always available. *Operator Mode(2) and SDK: Stall torque is only available when 'Torque Limit Mode' is off. Caution!! Using 'Torque Limit Mode' off in Operator Mode or using stall torque in Developer Mode may cause a rapid increase in product temperature and potential product damage. Please note that repairs (A/S) may be difficult in case of failure. (1) Chapter 5 Developer Mode ref. (2) Chapter 6 Operator Mode ref.	2
6	Rated Torque of Each Joint [Nm]	0.4
7	Absolute Encoder Resolution of Each Joint [bit, deg]	12, 0.1
8	Repeatability of Each Joint [deg]	0.1

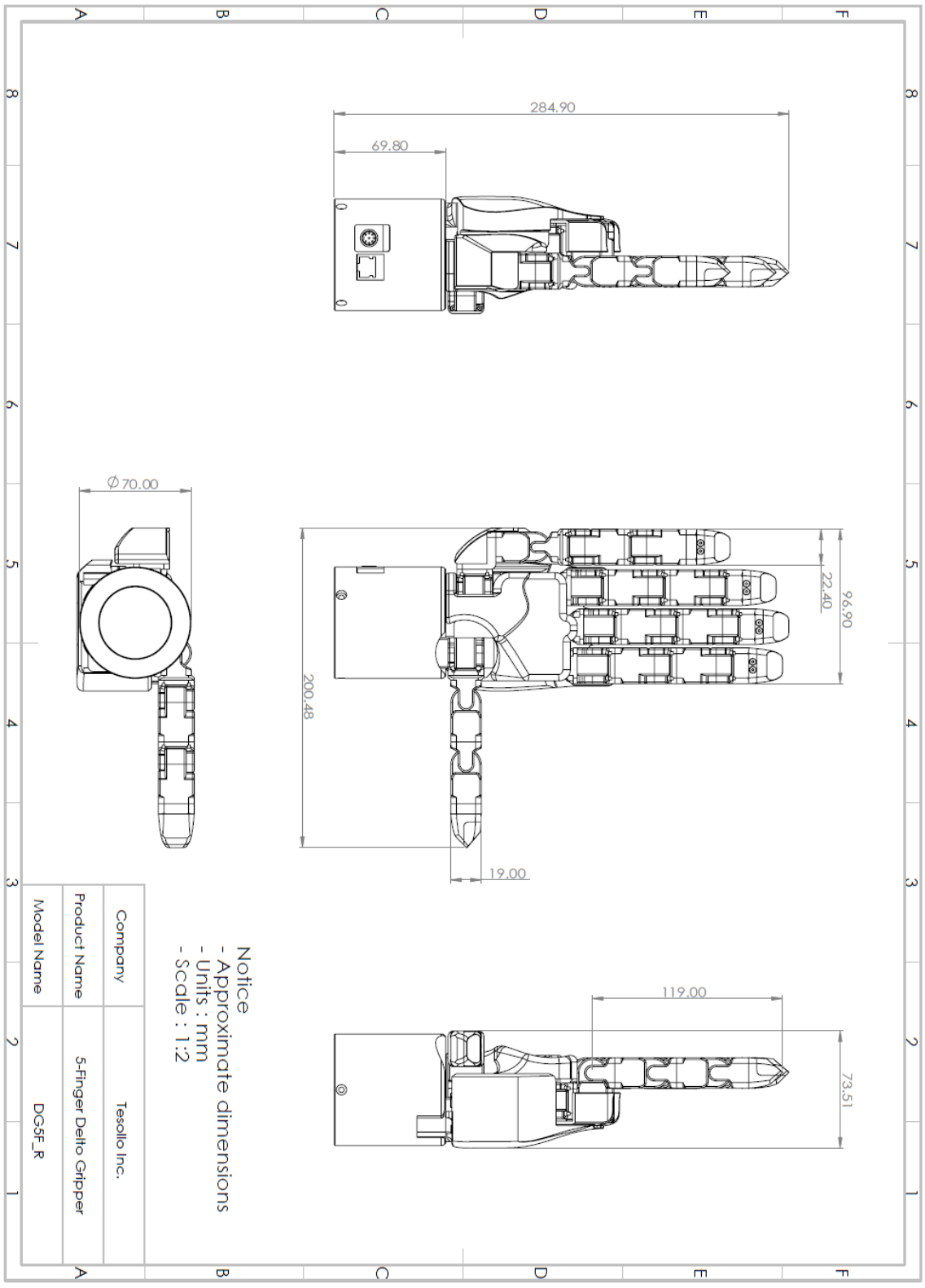
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Communication		
1	Communication Protocol	Modbus(RTU, TCP) EtherNET(TCP/IP)
2	Control Cycle [Hz]	250
3	Operator Mode Communication Cycle (RTU , TCP[3]) [Hz] *Varies depending on the control environment.	33 , 800[3]
4	Developer Mode Communication Cycle (EtherNET(TCP/IP) [3]) [Hz] *Varies depending on the control environment. (3) Measurement environment for communication cycle * OS : Windows 10 * CPU : AMD Ryzen 5 5600 6-Core Processor * RAM : 16GB	500[3]
Other Information		
1	Warranty Period [Months]	12

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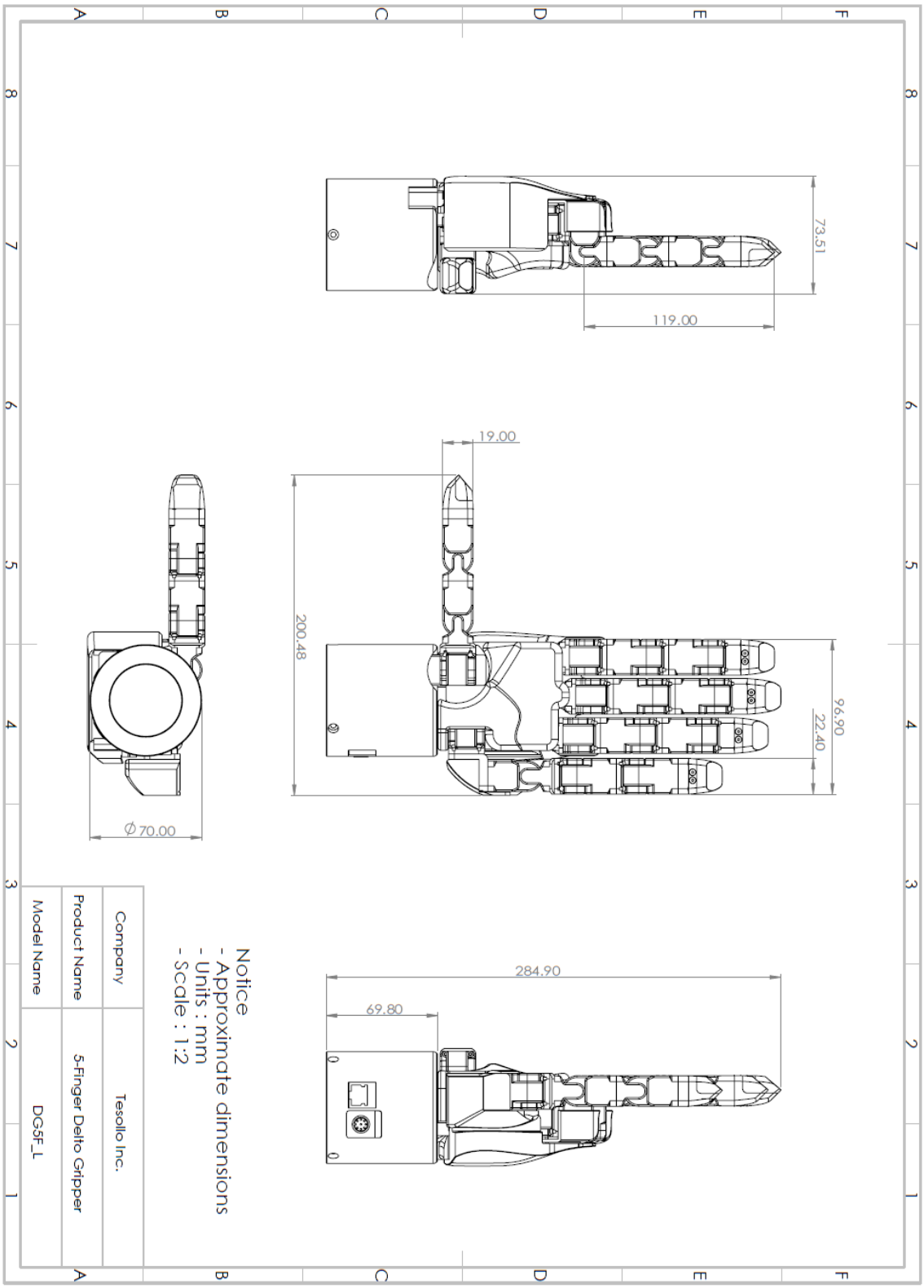
3.1.1 DG-5F dimensions

DG-5F Right hand



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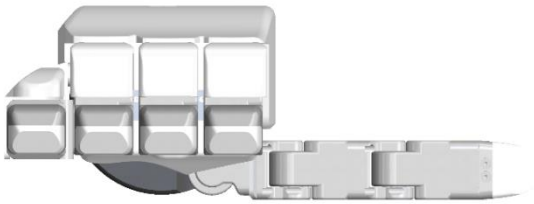
DG-5F Left hand

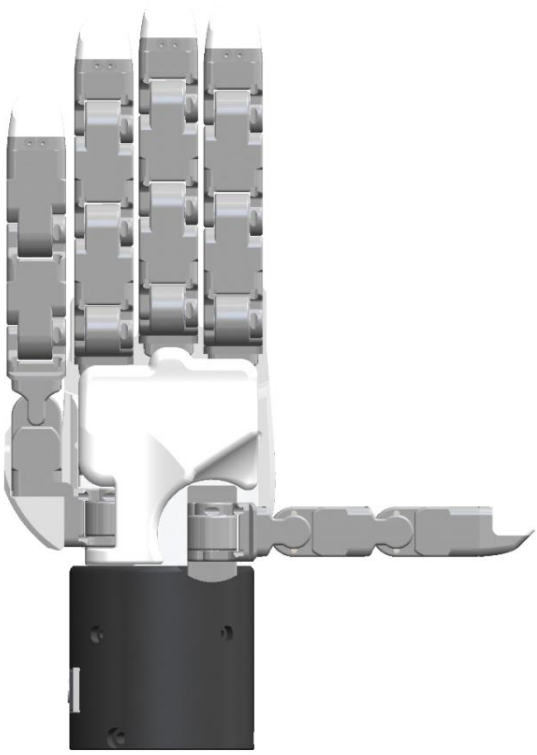


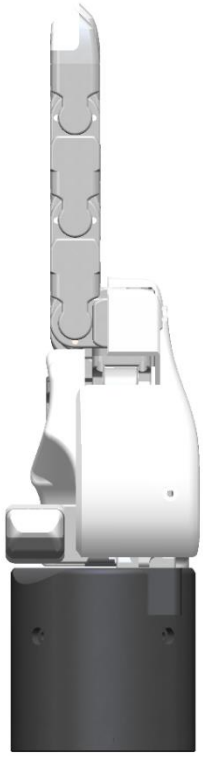
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3.1.2 DG-5F Zero Position

DG-5F Right hand

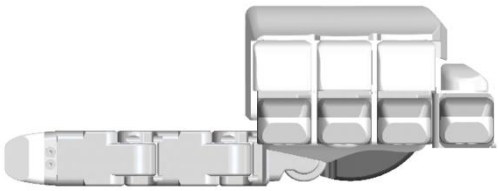


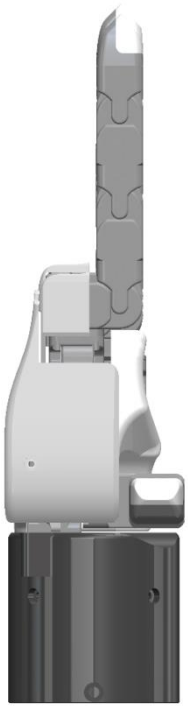


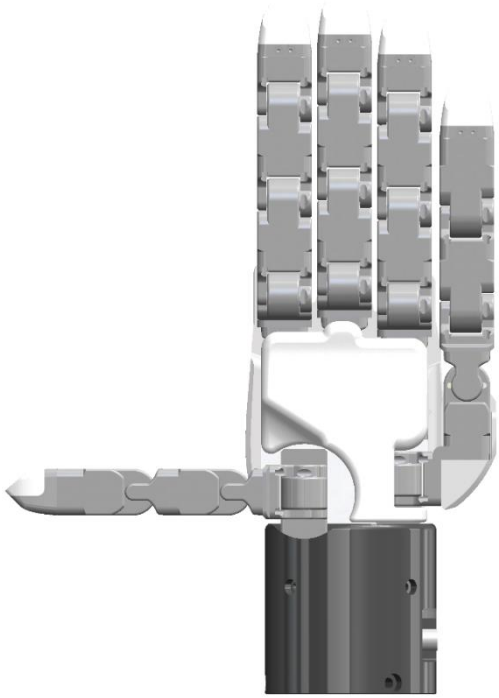


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DG-5F Left hand







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3.2 Technical Information of DG-5F

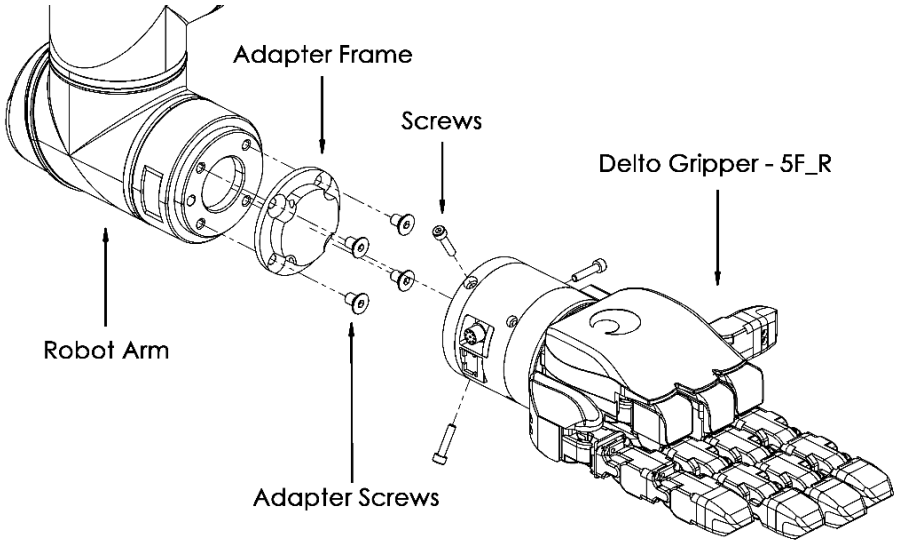
3.2.1 How to link

① Attach the provided adapter frame to the end of the collaborative robot.

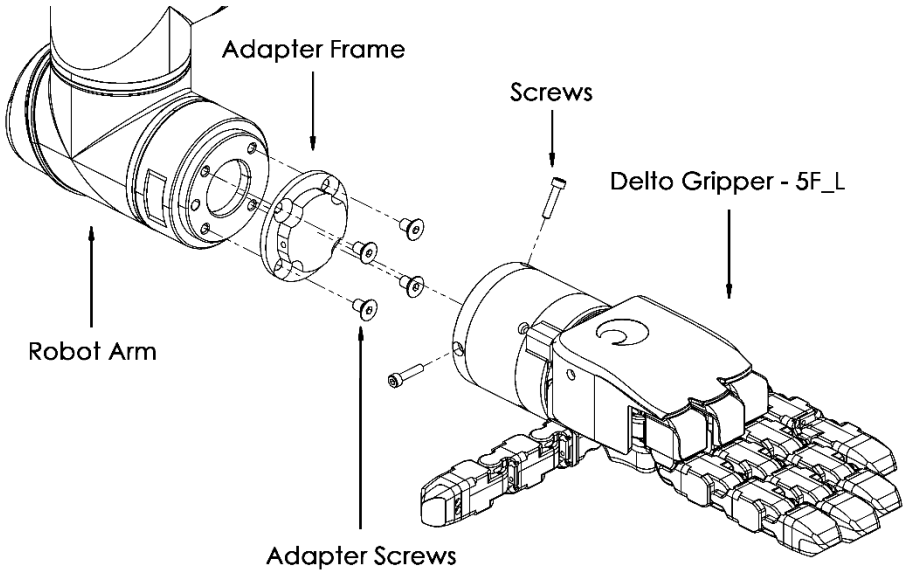
※ Standard : ISO 9409-1-50-4-M6

② Fix the adapter frame and DG-5F as shown in the figure below.

DG-5F Right hand

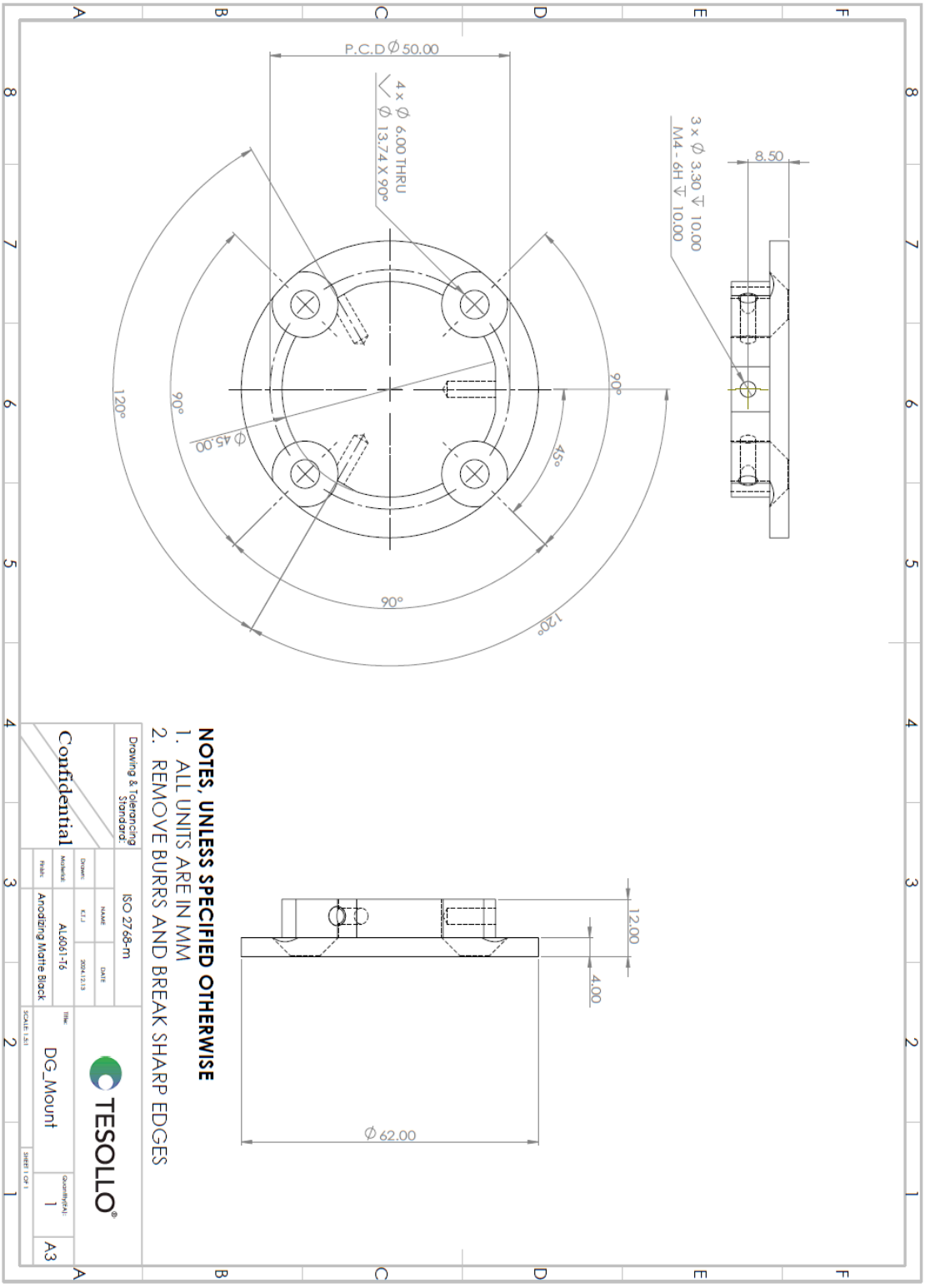


DG-5F Left hand



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3.2.2 Adapter frame dimensions

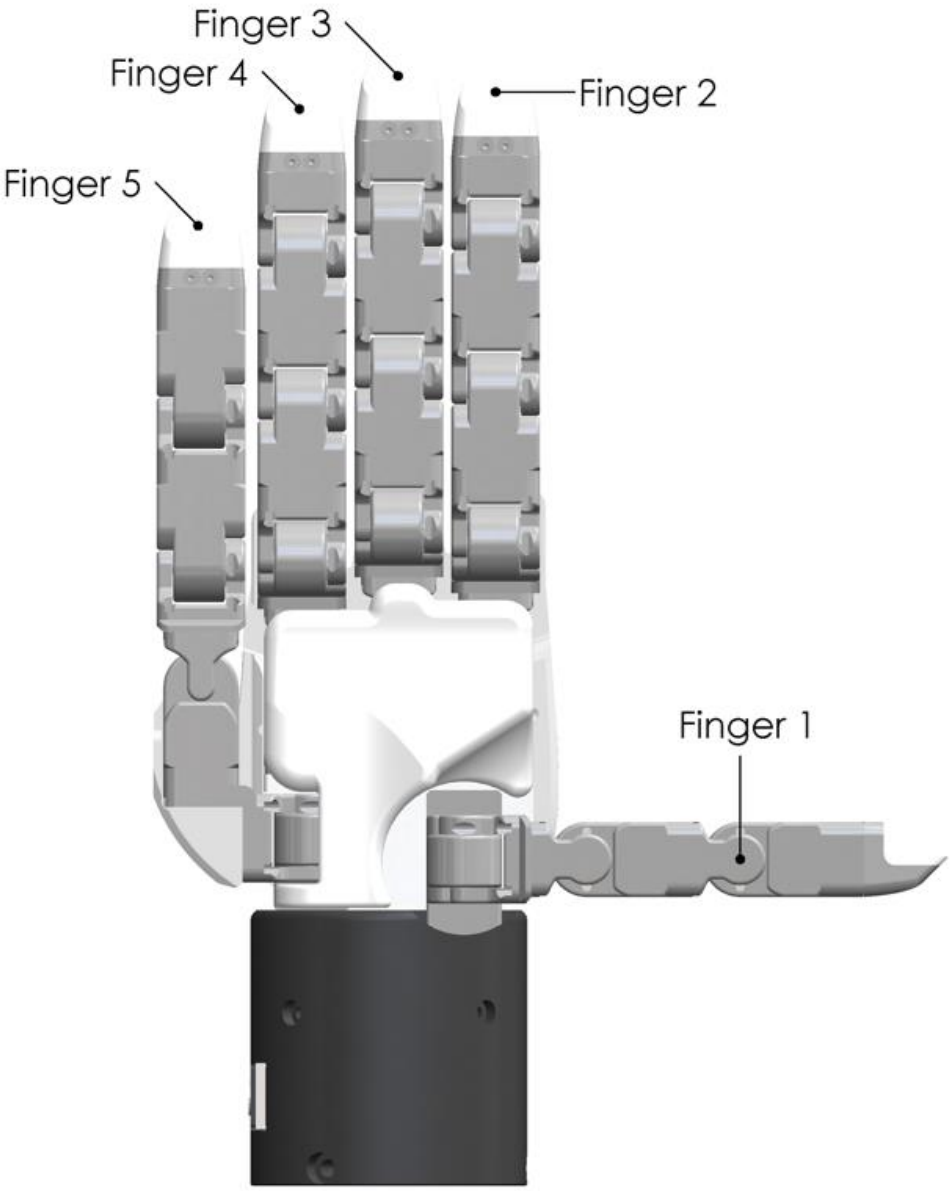


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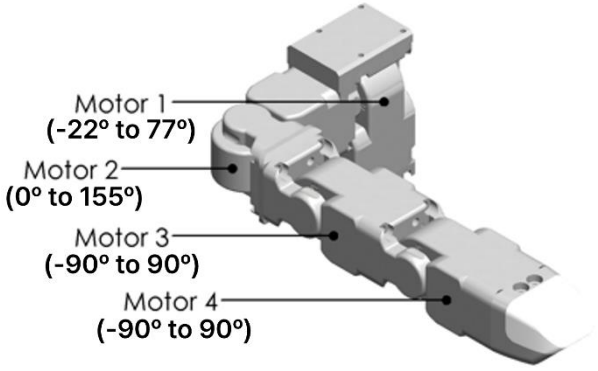
3.3 Basic Information about Motors

3.3.1 Motor sequence & Range of Motion

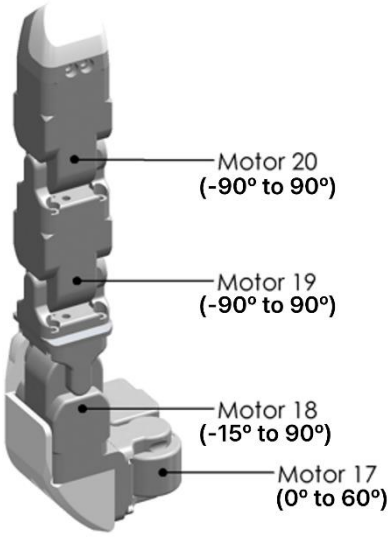
DG-5F Right hand



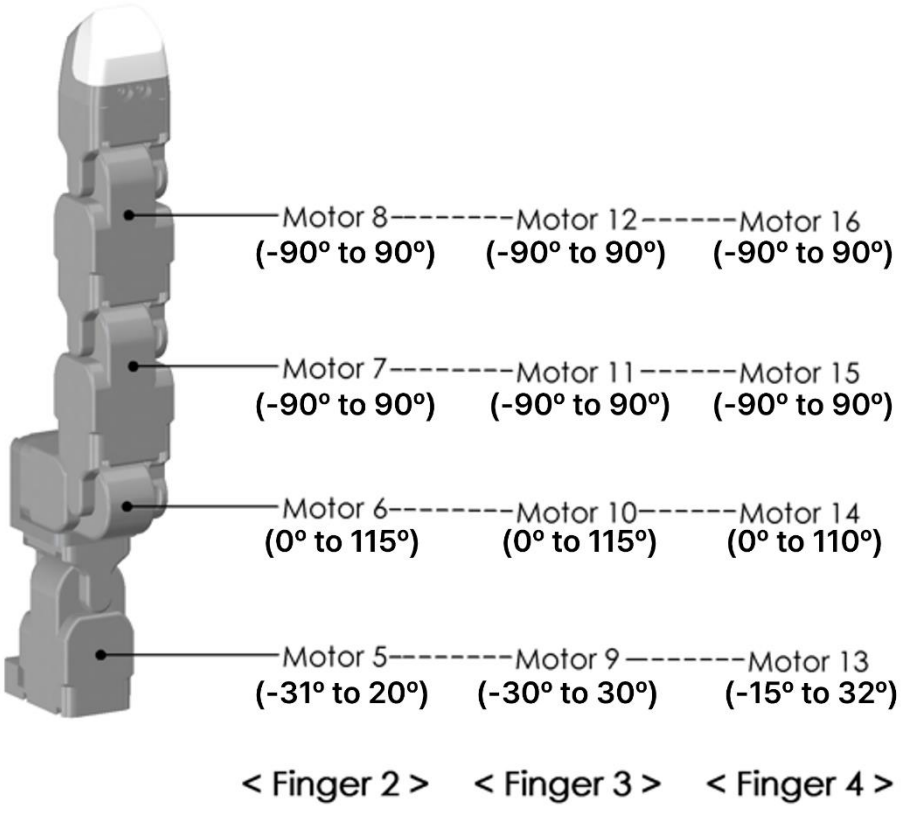
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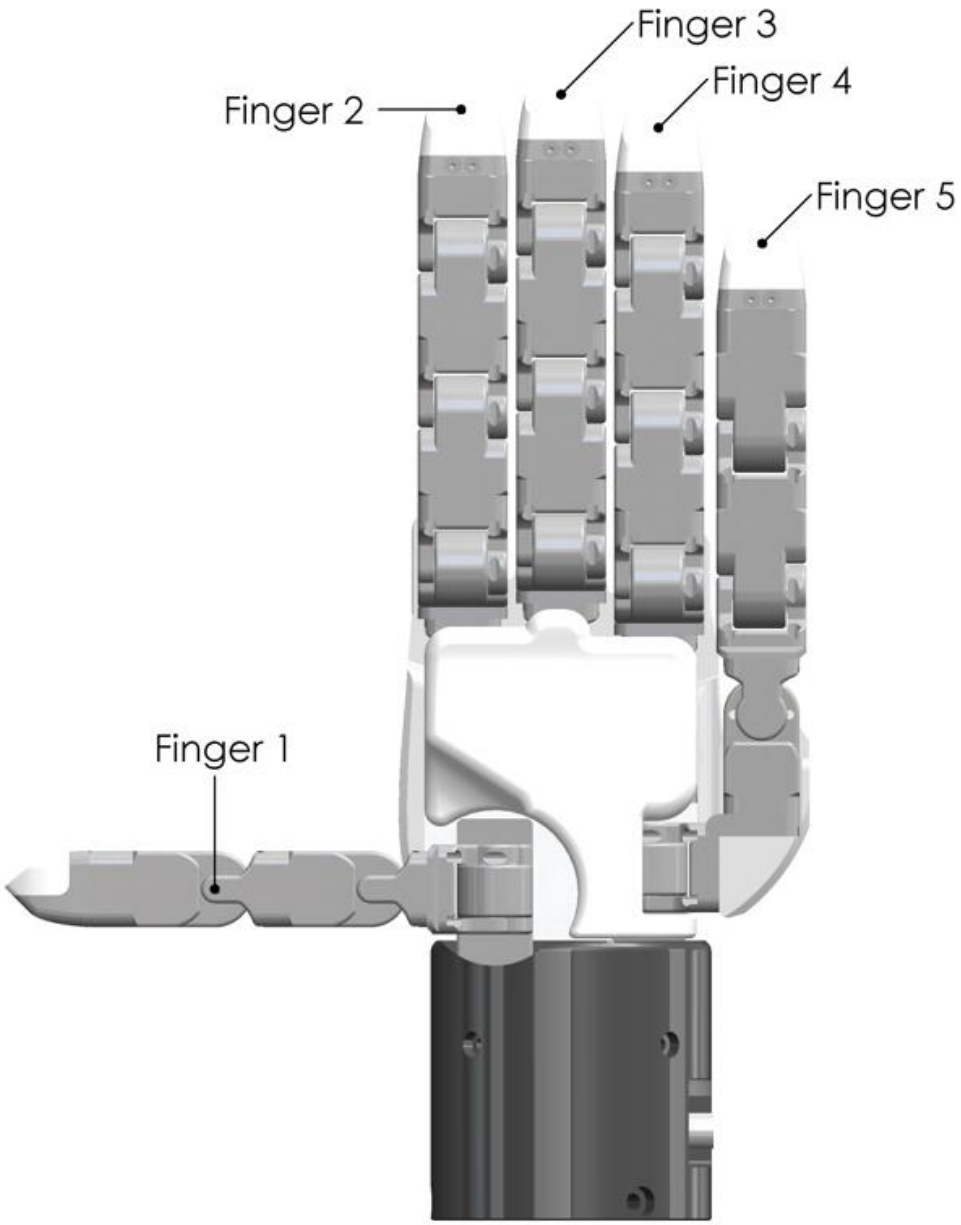
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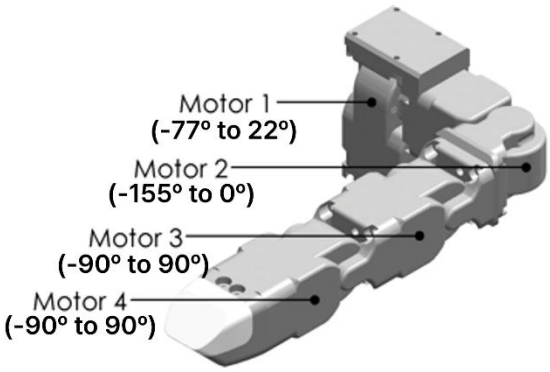
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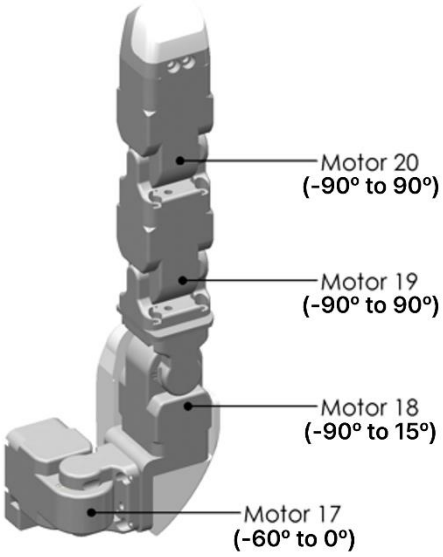
DG-5F Left hand



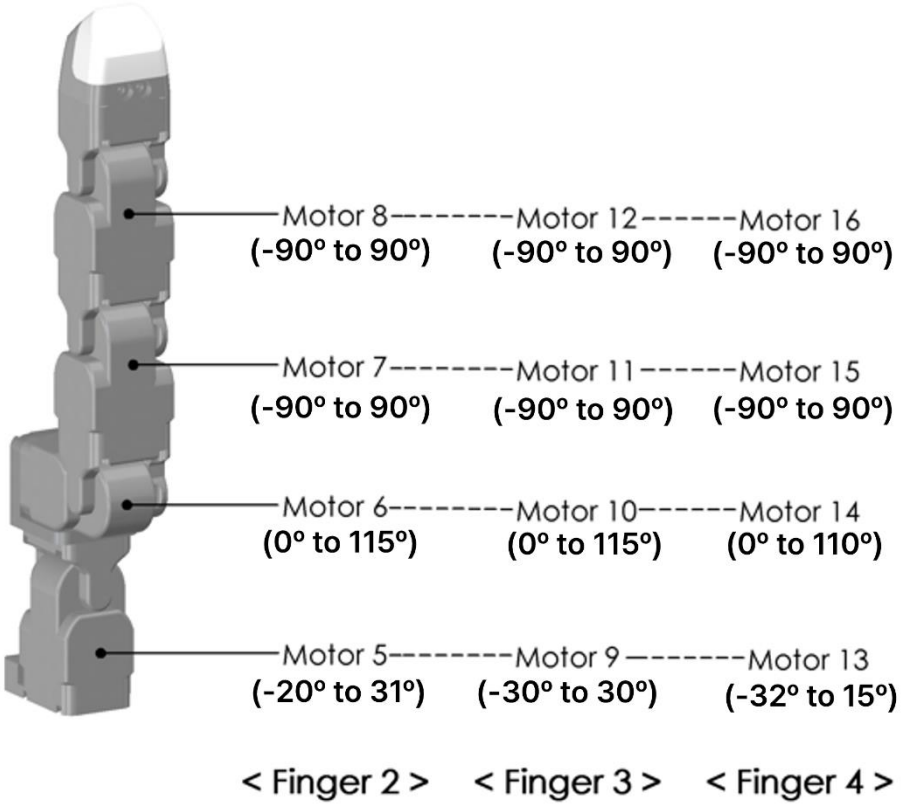
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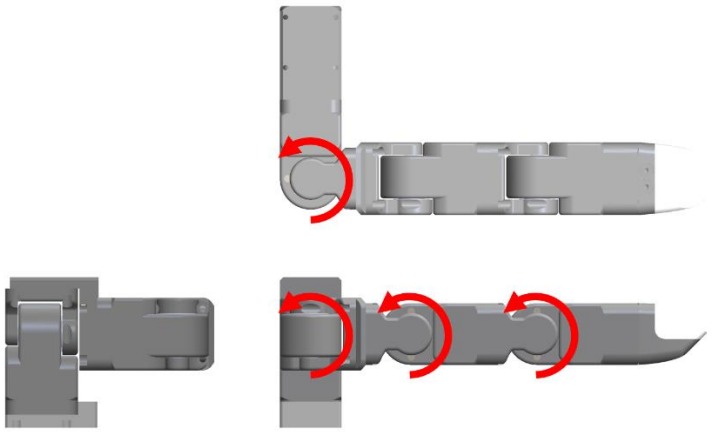


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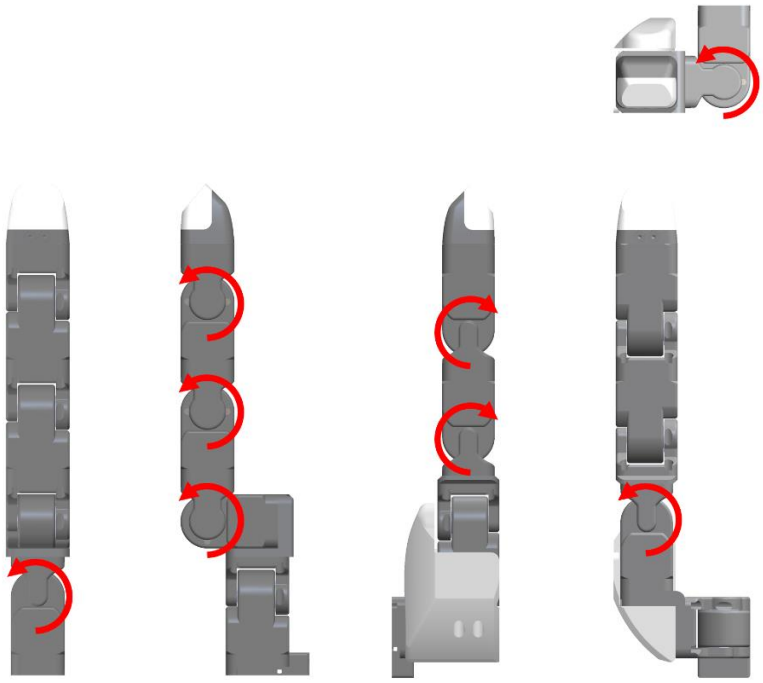
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3.3.2 Motor directions

DG-5F Right hand



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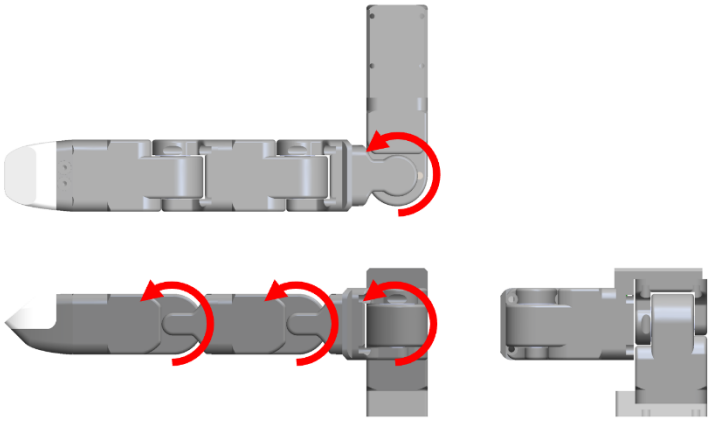
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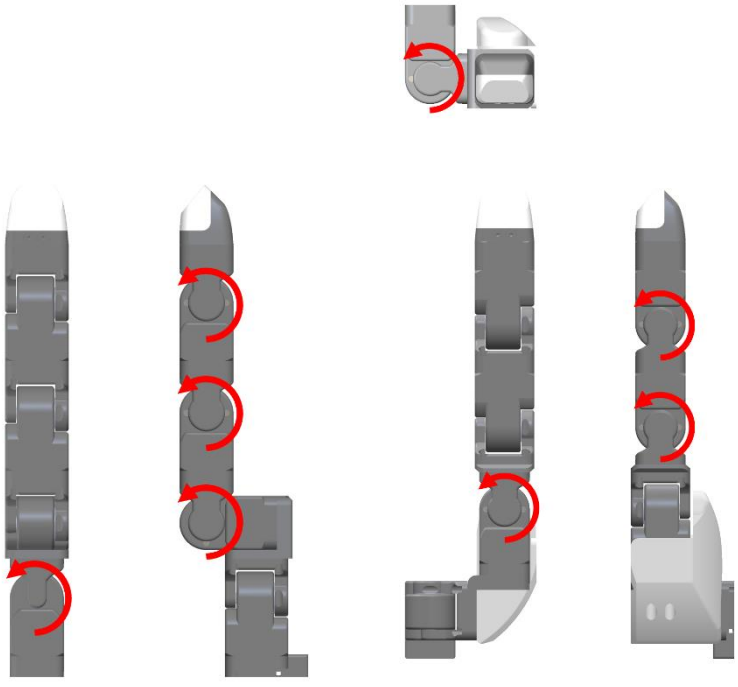
* The direction of the arrow is the positive direction.

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DG-5F Left hand



< Finger 1 >



< Finger 2~4 >

< Finger 5 >

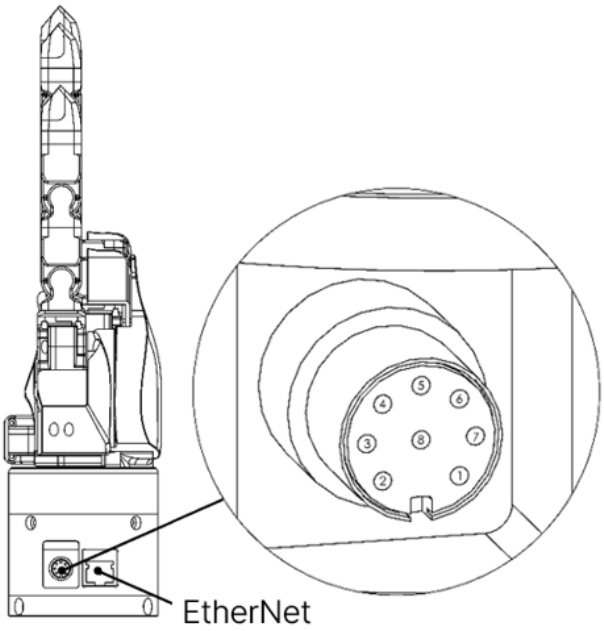
* The direction of the arrow is the positive direction.

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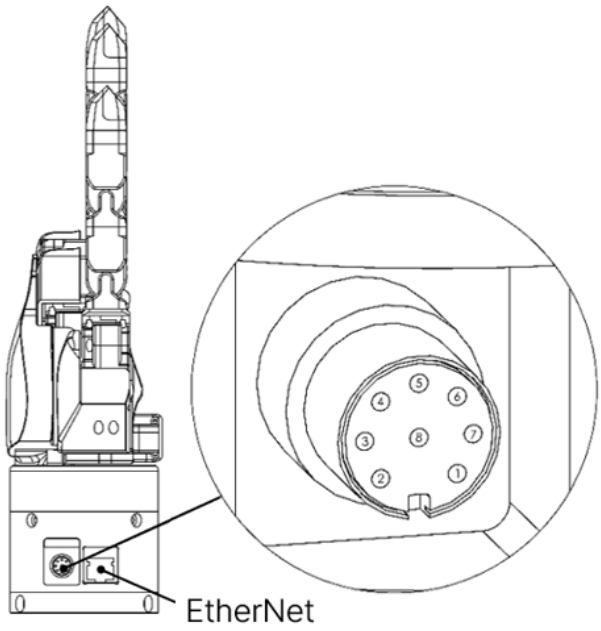
Chapter 4. Connector, Switch, LED Information

4.1 Connector Information

- DG-5F Left Hand



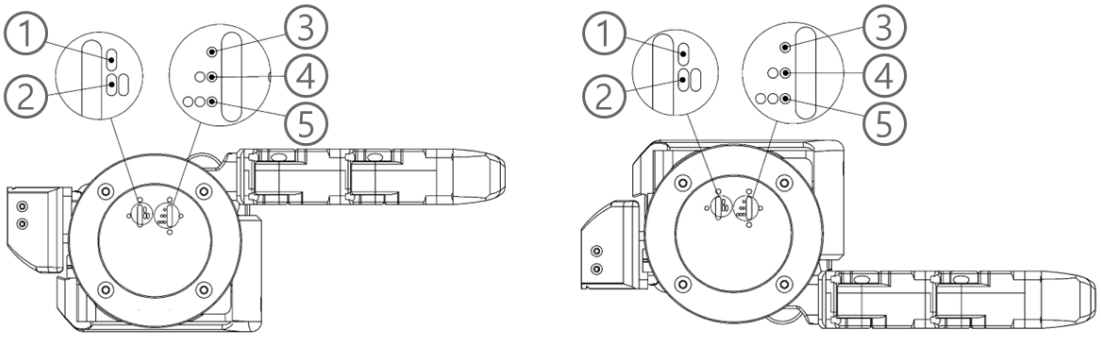
- DG-5F Right Hand



No.	Color	Connection
1	White	RS485 A
2	Brown	RS485 B
3	Green	Digital Output 1
4	Yellow	Digital Output 0
5	Gray	GND
6	Pink	Digital Output 2
7	Blue	Digital Input 0
8	Red	24V DC
* Power supply above 10A is required. * Input power DC 24V		

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4.2 Switch, LED Information



No.	System Modes	
①	Operator Mode	🔴🟡🔵 x 1 [All LEDs Blink Once]
②	Developer Mode	🔴🟡🔵 x 2 [All LEDs Blink Twice]
No.	**Communication Modes**	
③	RS485	○🟡○ [Yellow LED is On]
④	EtherNET	○○🔵 Socket Connect : Blue LED is On Socket Disconnect : Blue LED Blinks
⑤	Not used	Not used

① **Operator Mode:** This mode uses the internal controller of the product for operation. It supports the Modbus Protocol, allowing users to easily control the robotic hand by simply modifying the register map or using the provided UI.

② **Developer Mode:** This mode allows the user to control the robotic hand using a custom-developed controller. It enables the reception of information from each joint of the robotic hand, allowing direct control of each joint.

* When power is first applied, the control mode LED blinks, followed by the operation of the communication mode LED.
* Developer Mode is only available via Ethernet communication.
* If there is an issue with switch operation or communication, the red LED blinks.

*** Operator Mode: stall torque only available with 'Torque limit mode' off**
(6.2.1 Holding Register (Read/Write) 443 ref.)
*** Developer Mode: Always use stall torque**

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Chapter 5. Developer Mode

Developer Mode is a mode in which the user operates the system using a custom-developed controller. This mode allows direct control of each joint of the robot hand by receiving joint information.

Developer Mode supports communication via Ethernet (TCP/IP). This chapter introduces the protocol required for using Developer Mode.

Caution!!

Using 'Torque Limit Mode' off in Operator Mode or using stall torque in Developer Mode may cause a rapid increase in product temperature and potential product damage. Please note that repairs (A/S) may be difficult in case of failure.

5.1 Basic connection information

	DG-5F Right Hand	DG-5F Left Hand
IP address(1)	169.254.186.72	169.254.186.73
Subnet mask	255.255.255.0	
Gateway	0.0.0.0	
Port number(1)	502	
(1) It can be changed through the '5.3.4 Set IP & PORT (0x07)'		

5.2 Data Structure

5.2.1 Developer Mode Commands

Command	Name	Description
0x01	Get Data	Receive data from each joint
0x05	Set Duty	Sending Duty for each joint
0x06	Set GPIO	Transmit GPIO Output
0x07	Set IP & PORT	Gripper IP & PORT Settings
0x08	Get Product ID & Firmware Version	Receive Gripper Device ID & Firmware Version
0x09	Get Module Communication Error	Receive the Communication Anomaly Module ID
0x0A	Set Boot Mode	Enter boot mode
0x0B	Set F/T Sensor Offset	F/T Sensor Offset Setting

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5.2.2 Types of incoming data

Code	Name	Data size (Byte)	Remarks
0x01	Position	2	• **Unit**: 0.1 Degree • **Digits**: 1 decimal place → Data x 10 representation • **Byte order**: Big Endian
0x02	Current	2	• **Unit**: 1 mA • **Byte order**: Big Endian
0x03	Temperature	2	• **Unit**: 0.1 C° • **Digits**: 1 decimal place → Data x 10 representation • **Byte order**: Big Endian
0x04	RPM	1	• **Unit**: 1 RPM
0x05	F/T Sensor	12	• **Units**: Force[0.1 N] , Torque[0.1 Nm] • **Digits**: 1 decimal place → Data x 10 representation • **Byte order**: Big Endian • **Data order**: Fx, Fy, Fz, Tx, Ty, Tz [F = Force, T = Torque]
0x06	GPIO	4	• **Output Ports**: GPIO Output 1 - 3 • **Input Port**: GPIO Input 1 • **I/O levels**: 0 = 0V, 1 = 24V / 24mA

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5.3 Protocol Structure

5.3.1 Get Data(0x01)

The motor position value has a value of 0° at the robot's ' 3.1.2 DG-5F Zero Position ', and increases (+) when rotating the robot in the '3.3.2 Motor directions', and decreases (-) when rotating in the reverse direction.

Outgoing packets

- Send a command by including the data you want to receive in the Data area.

Length	CMD	Data 1	Data 2	Data 3	...
2 Byte	1 Byte	1 Byte	1 Byte	1 Byte	...

* **Length**: The total number of bytes in the packet, including the Length byte.
* **CMD**: 0x01 = Command to receive data from each joint
* **Data**: '5.2.2 Types of incoming data' ref
* **Example**: Receive position, current, and GPIO data → 00 06 01 01 02 06

Return Packets

- Receive data according to the type of transmitted data.

Length	CMD	ID 1	Data 1 to n	...	F/T Sensor	GPIO
2 Byte	1 Byte	1 Byte	1-2 Byte	...	12 Byte x m	4 Byte

* **Length**: The total number of bytes in the packet, including the Length byte.
* **CMD**: 0x01 = Command to receive data from each joint
* **ID + Data 1-n**: ID and Data for each joint (up to 20 = number of joints)
  - ID: ID of each joint
  - Data 1 through n: Sends user-specified data of position, current, temperature, or velocity
  - Data is received in the structure of [ID 1 + Data 1 + ... + Data n] for each joint
* **F/T Sensor**: When user selects F/T sensor
  - Transmit data in the following order: Fx, Fy, Fz, Tx, Ty, Tz
  - m : Number of fingers
  - Receive data according to the number of F/T sensors
  - ' 5.2.2 Types of incoming data '→ Refer to F/T Sensor
* **GPIO:** When the user selects a GPIO, it sends data in the following order
  - Receive data in the order: GPIO Output 1 – 3, then GPIO Input 1
  - GPIO Output: Returns the status value set by ' 5.3.3 Set GPIO (0x06) '
  - GPIO Input: Output the input level
  - ' 5.2.2 Types of incoming data ' → GPIO Reference

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5.3.2 Set Duty (0x05)

This command sends a Duty value to each joint to directly control each joint.

Outgoing packets

Length	CMD	ID 1	Data 1	ID 2	Data 2	...
2 Byte	1 Byte	1 Byte	2 Byte	1 Byte	2 Byte	...

* **Length**: The total number of bytes in the packet, including the Length byte.
* **CMD**: 0x05 = Duty send command for each joint
* **ID + Data 1-n**: ID and Data for each joint (up to 20 = number of joints)
  - ID: ID of each joint
  - Data: Duty of each joint. -1000 - +1000
    (1 decimal place → Data x 10 representation)
  - Transmit the data in the structure of [ID (1 Byte) + Data (2 Byte)] for each joint
  - Data byte order: Big Endian
  - Place iterations in a [ID(1 Byte) + Data(2 Bytes)] structure
* **Example**: All joints Duty 50% → 00 27 05 01 01 F4 02 01 F4 ... 0C 01 F4

5.3.3 Set GPIO (0x06)

Commands for outputting output to GPIO Output on the gripper.

Outgoing packets

Length	CMD	Data 1	Data 2	Data 3
2 Byte	1 Byte	1 Byte	1 Byte	1 Byte

* **Length**: The total number of bytes in the packet, including the Length byte.
* **CMD**: 0x06 = GPIO Output command
* **Data 1-3**: GPIO Output 1 - GPIO Output 3 Output Level Setting
  - Output Level: 0 = 0V, 1 = 24V / 24mA
* **Example**: GPIO Output 2 to 24V output → 00 06 06 00 01 00

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5.3.4 Set IP & PORT (0x07)

This command sets the IP address and PORT of the gripper.

Outgoing packets

Length	CMD	Data 1	Data 2
2 Byte	1 Byte	4 Byte	2 Byte

* **Length**: The total number of bytes in the packet, including the Length byte.
* **CMD**: 0x07 = Gripper IP & PORT Setup Command
* **Data 1**: Gripper IP address
  - Byte order: First through fourth digits of the IP
* **Data 2**: Gripper PORT
  - Byte order: Big Endian
* **Example**: IP = 169.254.186.72, PORT = 502 → 00 09 07 A9 FE BA 48 01 F6

5.3.5 Get Product ID & Firmware Version (0x08)

This command receives the Product ID and firmware version of the gripper.

Outgoing packets

Length	CMD
2 Byte	1 Byte

* **Length**: The total number of bytes in the packet, including the Length byte.
* **CMD**: 0x08 = Command to receive gripper Product ID & firmware version
* **Example**: 00 03 08

Return Packets

Length	CMD	Data 1	Data 2
2 Byte	1 Byte	2 Byte	2 Byte

* **Length**: The total number of bytes in the packet, including the Length byte.
* **CMD**: 0x08 = Command to receive gripper Product ID & firmware version
* **Data 1**: Product ID
  - DG-5F left hand = 5F 12 / DG-5F right hand = 5F 22
  - Byte order: Series, Model
* **Data 2**: Firmware Version
  - Byte order: Major, Minor
* **Example**: DG-5F left, firmware version 1.0 → 00 07 08 5F 12 01 00

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5.3.6 Get Module Communication Error (0x09)

This command detects communication errors for each joint.

Outgoing packets

Length	CMD
2 Byte	1 Byte
* **Length**: The total number of bytes in the packet, including the Length byte. * **CMD**: 0x09 = Command to receive communication anomaly module ID * **Example**: 00 03 09	

Return Packets

Length	CMD	Data
2 Byte	1 Byte	1 Byte
* **Length**: The total number of bytes in the packet, including the Length byte. * **CMD**: 0x09 = Command to receive communication anomaly module ID * **Data**: ID of the joint where the communication error occurred * **Example**: - Communication error on joint 2 → 00 04 09 02 - No error → 00 04 09 00		

5.3.7 Set Boot Mode (0x0A)

The command to enter boot mode for firmware download.

Outgoing packets

Length	CMD	Data
2 Byte	1 Byte	1 Byte
* **Length**: The total number of bytes in the packet, including the Length byte. * **CMD**: 0x0A = Command to enter boot mode * **Data**: Verification code = 0xAF * **Example**: 00 04 0A AF		

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5.3.8 Set F/T Sensor Offset (0x0B)

This command sets the offset of the F/T sensor.

Outgoing packets

Length	CMD
2 Byte	1 Byte
* **Length**: The total number of bytes in the packet, including the Length byte. * **CMD**: 0x0B = F/T Sensor Offset Setting Command * **Example**: 00 03 0B	

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Chapter 6. Operator Mode

This mode is controlled using the controller inside the product. It supports the Modbus Protocol and allows you to easily control the robot hand by simply changing the register map or by using the UI operations provided.

Caution!!

Using 'Torque Limit Mode' off in Operator Mode mode or using stall torque in Developer Mode may cause a rapid increase in product temperature and potential product damage. Please note that repairs (A/S) may be difficult in case of failure.

6.1 About Modbus basics

Supported Function	Read Holding Register (FC 03) Read Input Register (FC 04) Write Single Register (FC 06) Diagnostics (FC 08) Write Multiple Register (FC 16)
Slave ID(1)	1
Start Address	0
(1) Can be changed via Modbus Write (Holding Register 371) function	

6.1.1 Modbus RTU

Physical Interface	RS-485
Baudrate	115,200 bps
Data bits	8 bit
Parity bits	None
Stop bits	1 bit
Flow control	None
CRC	CRC-16 (Modbus)

6.1.2 Modbus TCP

	DG-5F Right Hand	**DG-5F Left Hand**
Physical Interface	EtherNET (TCP/IP)	
Ip Address(1)	169.254.186.72	169.254.186.73
Subnet mask(1)	255.255.255.0	
Gateway (1)	0.0.0.0	
Port number (1)	502	
(1) Can be changed via Modbus Write (Holding Register 372,376,380,384) function		

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6.2 Modbus Registers

*(x10, x100, x1000): Write/Read data multiplied by 10 to apply decimalization

* Shade of green: EEPROM area (Holding Reigister 1)

6.2.1 Holding Register (Read / Write)

Address	Description.	Scale	Initial values
DEC			DEC
0	**System Behavior [0 to 1]** • 0: Stop • 1: Start	-	0
1	**EEPROM Write [1]** • 1: Store data in EEPROM memory	-	0
2	**Motion Teach Mode [0 to 1]** • 0: Teach Mode Off • 1: Teach Mode On • Ability to physically shape the gripper yourself	-	0
3	**Load Pose Recipe [1 to 100]** • Importing a saved pose recipe	-	1
5	**Save Current Position Recipe [1 to 100]** • Store the current position value in slot n of the recipe	-	0
6	**Save Target Position Recipe [1 to 100]** • Store the target position value in slot n of the recipe	-	0
7	**Motor 1 Position Target Value [-1800 to 1800]** • Unit: 0.1°	x10	0
8	**Motor 2 Position Target Value [-1800 to 1800]**	x10	0
9	**Motor 3 Position Target Value [-1800 to 1800]**	x10	0
10	**Motor 4 Position Target Value [-1800 to 1800]**	x10	0
11	**Motor 5 Position Target Value [-1800 to 1800]**	x10	0
12	**Motor 6 Position Target Value [-1800 to 1800]**	x10	0
13	**Motor 7 Position Target Value [-1800 to 1800]**	x10	0
14	**Motor 8 Position Target Value [-1800 to 1800]**	x10	0

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15	**Motor 9 Position Target Value [-1800 to 1800]**	x10	0
16	**Motor 10 Position Target Value [-1800 to 1800]**	x10	0
17	**Motor 11 Position Target Value [-1800 to 1800]**	x10	0
18	**Motor 12 Position Target Value [-1800 to 1800]**	x10	0
19	**Motor 13 Position Target Value [-1800 to 1800]**	x10	0
20	**Motor 14 Position Target Value [-1800 to 1800]**	x10	0
21	**Motor 15 Position Target Value [-1800 to 1800]**	x10	0
22	**Motor 16 Position Target Value [-1800 to 1800]**	x10	0
23	**Motor 17 Position Target Value [-1800 to 1800]**	x10	0
24	**Motor 18 Position Target Value [-1800 to 1800]**	x10	0
25	**Motor 19 Position Target Value [-1800 to 1800]**	x10	0
26	**Motor 20 Position Target Value [-1800 to 1800]**	x10	0
27	**Motor 1 Target Position Reach Time [0 to 30000]** • Unit: 1 ms • Arrive at the target position on time	-	300
28	**Motor 2 Target Position Reach Time [0 to 30000]**	-	300
29	**Motor 3 Target Position Reach Time [0 to 30000]**	-	300
30	**Motor 4 Target Position Reach Time [0 to 30000]**	-	300
31	**Motor 5 Target Position Reach Time [0 to 30000]**	-	300
32	**Motor 6 Target Position Reach Time [0 to 30000]**	-	300
33	**Motor 7 Target Position Reach Time [0 to 30000]**	-	300
34	**Motor 8 Target Position Reach Time [0 to 30000]**	-	300
35	**Motor 9 Target Position Reach Time [0 to 30000]**	-	300
36	**Motor 10 Target Position Reach Time [0 to 30000]**	-	300
37	**Motor 11 Target Position Reach Time [0 to 30000]**	-	300
38	**Motor 12 Target Position Reach Time [0 to 30000]**	-	300
39	**Motor 13 Target Position Reach Time [0 to 30000]**	-	300
40	**Motor 14 Target Position Reach Time [0 to 30000]**	-	300
41	**Motor 15 Target Position Reach Time [0 to 30000]**	-	300
42	**Motor 16 Target Position Reach Time [0 to 30000]**	-	300
43	**Motor 17 Target Position Reach Time [0 to 30000]**	-	300
44	**Motor 18 Target Position Reach Time [0 to 30000]**	-	300
45	**Motor 19 Target Position Reach Time [0 to 30000]**	-	300
46	**Motor 20 Target Position Reach Time [0 to 30000]**	-	300

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82	**Load Gain Recipes [1 to 20]** • Importing a saved gain recipe • See Holding Register 83 for storing gain recipes.	-	1
83	**Save Gain Recipe [1 to 20]** • Save the P Gain, D Gain, I Gain, I Limit, and Gain Mode settings to slot n.	-	0
85	**Gain Mode [0 to 1]** • 0: PD control • 1: PID Control	-	0
86	**Motor 1 P Gain [0 to 30000]** • Unit: 0.1	x10	10
87	**Motor 2 P Gain [0 to 30000]**	x10	10
88	**Motor 3 P Gain [0 to 30000]**	x10	10
89	**Motor 4 P Gain [0 to 30000]**	x10	10
90	**Motor 5 P Gain [0 to 30000]**	x10	10
91	**Motor 6 P Gain [0 to 30000]**	x10	10
92	**Motor 7 P Gain [0 to 30000]**	x10	10
93	**Motor 8 P Gain [0 to 30000]**	x10	10
94	**Motor 9 P Gain [0 to 30000]**	x10	10
95	**Motor 10 P Gain [0 to 30000]**	x10	10
96	**Motor 11 P Gain [0 to 30000]**	x10	10
97	**Motor 12 P Gain [0 to 30000]**	x10	10
98	**Motor 13 P Gain [0 to 30000]**	x10	10
99	**Motor 14 P Gain [0 to 30000]**	x10	10
100	**Motor 15 P Gain [0 to 30000]**	x10	10
101	**Motor 16 P Gain [0 to 30000]**	x10	10
102	**Motor 17 P Gain [0 to 30000]**	x10	10
103	**Motor 18 P Gain [0 to 30000]**	x10	10
104	**Motor 19 P Gain [0 to 30000]**	x10	10
105	**Motor 20 P Gain [0 to 30000]**	x10	10
106	**Motor 1 D Gain [0 to 30000]** • Unit: 0.1	x10	30
107	**Motor 2 D Gain [0 to 30000]**	x10	30
108	**Motor 3 D Gain [0 to 30000]**	x10	30
109	**Motor 4 D Gain [0 to 30000]**	x10	30

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110	**Motor 5 D Gain [0 to 30000]**	x10	30
111	**Motor 6 D Gain [0 to 30000]**	x10	30
112	**Motor 7 D Gain [0 to 30000]**	x10	30
113	**Motor 8 D Gain [0 to 30000]**	x10	30
114	**Motor 9 D Gain [0 to 30000]**	x10	30
115	**Motor 10 D Gain [0 to 30000]**	x10	30
116	**Motor 11 D Gain [0 to 30000]**	x10	30
117	**Motor 12 D Gain [0 to 30000]**	x10	30
118	**Motor 13 D Gain [0 to 30000]**	x10	30
119	**Motor 14 D Gain [0 to 30000]**	x10	30
120	**Motor 15 D Gain [0 to 30000]**	x10	30
121	**Motor 16 D Gain [0 to 30000]**	x10	30
122	**Motor 17 D Gain [0 to 30000]**	x10	30
123	**Motor 18 D Gain [0 to 30000]**	x10	30
124	**Motor 19 D Gain [0 to 30000]**	x10	30
125	**Motor 20 D Gain [0 to 30000]**	x10	30
126	**Motor 1 I Gain [0 to 30000]** • Unit: 0.001	x1000	0
127	**Motor 2 I Gain [0 to 30000]**	x1000	0
128	**Motor 3 I Gain [0 to 30000]**	x1000	0
129	**Motor 4 I Gain [0 to 30000]**	x1000	0
130	**Motor 5 I Gain [0 to 30000]**	x1000	0
131	**Motor 6 I Gain [0 to 30000]**	x1000	0
132	**Motor 7 I Gain [0 to 30000]**	x1000	0
133	**Motor 8 I Gain [0 to 30000]**	x1000	0
134	**Motor 9 I Gain [0 to 30000]**	x1000	0
135	**Motor 10 I Gain [0 to 30000]**	x1000	0
136	**Motor 11 I Gain [0 to 30000]**	x1000	0
137	**Motor 12 I Gain [0 to 30000]**	x1000	0
138	**Motor 13 I Gain [0 to 30000]**	x1000	0
139	**Motor 14 I Gain [0 to 30000]**	x1000	0
140	**Motor 15 I Gain [0 to 30000]**	x1000	0
141	**Motor 16 I Gain [0 to 30000]**	x1000	0
142	**Motor 17 I Gain [0 to 30000]**	x1000	0
143	**Motor 18 I Gain [0 to 30000]**	x1000	0

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144	**Motor 19 I Gain [0 to 30000]**	x1000	0
145	**Motor 20 I Gain [0 to 30000]**	x1000	0
146	**Motor 1 I Limit [0 to 30000]** • Unit: 0.01	x100	0
147	**Motor 2 I Limit [0 to 30000]**	x100	0
148	**Motor 3 I Limit [0 to 30000]**	x100	0
149	**Motor 4 I Limit [0 to 30000]**	x100	0
150	**Motor 5 I Limit [0 to 30000]**	x100	0
151	**Motor 6 I Limit [0 to 30000]**	x100	0
152	**Motor 7 I Limit [0 to 30000]**	x100	0
153	**Motor 8 I Limit [0 to 30000]**	x100	0
154	**Motor 9 I Limit [0 to 30000]**	x100	0
155	**Motor 10 I Limit [0 to 30000]**	x100	0
156	**Motor 11 I Limit [0 to 30000]**	x100	0
157	**Motor 12 I Limit [0 to 30000]**	x100	0
158	**Motor 13 I Limit [0 to 30000]**	x100	0
159	**Motor 14 I Limit [0 to 30000]**	x100	0
160	**Motor 15 I Limit [0 to 30000]**	x100	0
161	**Motor 16 I Limit [0 to 30000]**	x100	0
162	**Motor 17 I Limit [0 to 30000]**	x100	0
163	**Motor 18 I Limit [0 to 30000]**	x100	0
164	**Motor 19 I Limit [0 to 30000]**	x100	0
165	**Motor 20 I Limit [0 to 30000]**	x100	0
286	**Start Blend Joint [0 to 10]** • 0: Stop • 1-10: Start the Blend Joint saved in n. • Execute the recipes stored in the blend list sequentially.	-	0
287	**Save The Current Blend List [1 to 10]** Save the Blend List to slot n	-	0
288	**Add and Delete Blend Lists [0 to 2]** • Add Pose Recipes, Gain Recipes, and Latency to the Blend List (up to 50) • 0: default • 1: Add to list • 2: Initialize the list	-	0

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289	**Select a Blend Pose Recipe [1 to 100]** • Choose from your saved pose recipes	-	1
290	**Select a Blend Gain Recipe [1 to 20]** • Choose from your saved gain recipes	-	1
291	**Blend Target Position Wait Time[0 to 30000]** • Unit: 1 ms • Setting the wait time between each action during Blend actions	-	0
292	**Grasp Action [0 to 1]** • 0: RELEASE • 1: GRASP	-	0
293	**Load the Grasp Recipe [1 to 20]** • Recalling saved grasping settings	-	1
294	**Save the Grasp Recipe [1 to 20]** • Save “Grasp Mode”, “Grasp Force”, “Hold Position” to slot n	-	0
295	**Grasp Mode [21 to 29]** • 6.2.3 Grasp Mode ref	-	21
296	**Grasp Force [0 to 200]** • Unit: 0.1 N	-	30
299	**Motor 1 Hold Position During Grasping [0 to 1]** • 0: Off • 1: Lock position	-	0
300	**Motor 2 Hold Position During Grasping [0 to 1]**	-	0
301	**Motor 3 Hold Position During Grasping [0 to 1]**	-	0
302	**Motor 4 Hold Position During Grasping [0 to 1]**	-	0
303	**Motor 5 Hold Position During Grasping [0 to 1]**	-	0
304	**Motor 6 Hold Position During Grasping [0 to 1]**	-	0
305	**Motor 7 Hold Position During Grasping [0 to 1]**	-	0
306	**Motor 8 Hold Position During Grasping [0 to 1]**	-	0
307	**Motor 9 Hold Position During Grasping [0 to 1]**	-	0
308	**Motor 10 Hold Position During Grasping [0 to 1]**	-	0

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309	**Motor 11 Hold Position During Grasping [0 to 1]**	-	0
310	**Motor 12 Hold Position During Grasping [0 to 1]**	-	0
311	**Motor 13 Hold Position During Grasping [0 to 1]**	-	0
312	**Motor 14 Hold Position During Grasping [0 to 1]**	-	0
313	**Motor 15 Hold Position During Grasping [0 to 1]**	-	0
314	**Motor 16 Hold Position During Grasping [0 to 1]**	-	0
315	**Motor 17 Hold Position During Grasping [0 to 1]**	-	0
316	**Motor 18 Hold Position During Grasping [0 to 1]**	-	0
317	**Motor 19 Hold Position During Grasping [0 to 1]**	-	0
318	**Motor 20 Hold Position During Grasping [0 to 1]**	-	0
319	**Motor 1 Offset [-1800 to 1800]** • Unit: 0.1° • Setting Offsets for Joint Angle Values	x10	0
320	**Motor 2 Offset [-1800 to 1800]**	x10	0
321	**Motor 3 Offset [-1800 to 1800]**	x10	0
322	**Motor 4 Offset [-1800 to 1800]**	x10	0
323	**Motor 5 Offset [-1800 to 1800]**	x10	0
324	**Motor 6 Offset [-1800 to 1800]**	x10	0
325	**Motor 7 Offset [-1800 to 1800]**	x10	0
326	**Motor 8 Offset [-1800 to 1800]**	x10	0
327	**Motor 9 Offset [-1800 to 1800]**	x10	0
328	**Motor 10 Offset [-1800 to 1800]**	x10	0
329	**Motor 11 Offset [-1800 to 1800]**	x10	0
330	**Motor 12 Offset [-1800 to 1800]**	x10	0
331	**Motor 13 Offset [-1800 to 1800]**	x10	0
332	**Motor 14 Offset [-1800 to 1800]**	x10	0
333	**Motor 15 Offset [-1800 to 1800]**	x10	0
334	**Motor 16 Offset [-1800 to 1800]**	x10	0
335	**Motor 17 Offset [-1800 to 1800]**	x10	0
336	**Motor 18 Offset [-1800 to 1800]**	x10	0

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337	**Motor 19 Offset [-1800 to 1800]**	x10	0
338	**Motor 20 Offset [-1800 to 1800]**	x10	0
339	**Motor 1 Inpose [0 to 1800]** • Unit: 0.1° • Range for determining if the target position has been reached • Input Register 3 ref	x10	100
340	**Motor 2 Inpose [0 to 1800]**	x10	100
341	**Motor 3 Inpose [0 to 1800]**	x10	100
342	**Motor 4 Inpose [0 to 1800]**	x10	100
343	**Motor 5 Inpose [0 to 1800]**	x10	100
344	**Motor 6 Inpose [0 to 1800]**	x10	100
345	**Motor 7 Inpose [0 to 1800]**	x10	100
346	**Motor 8 Inpose [0 to 1800]**	x10	100
347	**Motor 9 Inpose [0 to 1800]**	x10	100
348	**Motor 10 Inpose [0 to 1800]**	x10	100
349	**Motor 11 Inpose [0 to 1800]**	x10	100
350	**Motor 12 Inpose [0 to 1800]**	x10	100
351	**Motor 13 Inpose [0 to 1800]**	x10	100
352	**Motor 14 Inpose [0 to 1800]**	x10	100
353	**Motor 15 Inpose [0 to 1800]**	x10	100
354	**Motor 16 Inpose [0 to 1800]**	x10	100
355	**Motor 17 Inpose [0 to 1800]**	x10	100
356	**Motor 18 Inpose [0 to 1800]**	x10	100
357	**Motor 19 Inpose [0 to 1800]**	x10	100
358	**Motor 20 Inpose [0 to 1800]**	x10	100
369	**Moving Inpose [0 to 1800]** • Range of whether to determine gripper movement • Unit: 0.1° • Input Register 2 ref	x10	3
371	**Modbus Slave ID [1 to 255]**	-	1
372	**EtherNET IP 1 [1 to 255]**	-	169
373	**EtherNET IP 2 [0 to 255]**	-	254
374	**EtherNET IP 3 [0 to 255]**	-	186
375	**EtherNET IP 4 [0 to 255]**	-	72or73

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376	**EtherNET subnet 1 [0 to 255]**	-	255
377	**EtherNET subnet 2 [0 to 255]**	-	255
378	**EtherNET subnet 3 [0 to 255]**	-	255
379	**EtherNET subnet 4 [0 to 255]**	-	0
380	**EtherNET Gateway 1 [0 to 255]**	-	0
381	**EtherNET Gateway 2 [0 to 255]**	-	0
382	**EtherNET Gateway 3 [0 to 255]**	-	0
383	**EtherNET Gateway 4 [0 to 255]**	-	0
384	**EtherNET Port [1 to 30000]** • The value 2607 cannot be entered.	-	502
385	**Set the initial Pose Recipe [1 to 100]** • Set which of your saved pose recipes to apply on first launch	-	1
386	**Set the initial Gain Recipe [1 to 20]** • Set the gain to apply on first startup from a saved gain recipe	-	1
387	**Set the initial Grasp Recipe [1 to 20]** • Which of your saved grasp settings to apply on first startup	-	1
388	**Set the Low Pass Filter Alpha Value [0 to 99]** • Apply a Low Pass Filter to the Joint Current Position Value • 0: Off • 1 to 99: Set the Alpha value (%)	-	0
389	**Set Current Control Mode [0 to 1]** • 0: Off • 1: On • P, I, and D gain values cannot be set in current control mode.	-	0
390	**Motor 1 Current Target Value [-150 to 150]** • Torque Limit Mode off [-1300 to 1300] • Unit: 1 mA	-	0
391	**Motor 2 Current Target Value [-150 to 150]**	-	0
392	**Motor 3 Current Target Value [-150 to 150]**	-	0
393	**Motor 4 Current Target Value [-150 to 150]**	-	0
394	**Motor 5 Current Target Value [-150 to 150]**	-	0
395	**Motor 6 Current Target Value [-150 to 150]**	-	0
396	**Motor 7 Current Target Value [-150 to 150]**	-	0

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397	**Motor 8 Current Target Value [-150 to 150]**	-	0
398	**Motor 9 Current Target Value [-150 to 150]**	-	0
399	**Motor 10 Current Target Value [-150 to 150]**	-	0
400	**Motor 11 Current Target Value [-150 to 150]**	-	0
401	**Motor 12 Current Target Value [-150 to 150]**	-	0
402	**Motor 13 Current Target Value [-150 to 150]**	-	0
403	**Motor 14 Current Target Value [-150 to 150]**	-	0
404	**Motor 15 Current Target Value [-150 to 150]**	-	0
405	**Motor 16 Current Target Value [-150 to 150]**	-	0
406	**Motor 17 Current Target Value [-150 to 150]**	-	0
407	**Motor 18 Current Target Value [-150 to 150]**	-	0
408	**Motor 19 Current Target Value [-150 to 150]**	-	0
409	**Motor 20 Current Target Value [-150 to 150]**	-	0
439	**GPIO Output 1 Level [0 to 1]** • 0: 0 V • 1: 24 V / 24mA	-	0
440	**GPIO Output 2 Level [0 to 1]**	-	0
441	**GPIO Output 3 Level [0 to 1]**	-	0
442	**Enter Boot Mode [2607]** • Enter boot mode for firmware update	-	0
443	**Torque Limit Mode [0 to 1]** • 0: Torque Limit off • 1: Torque Limit on	-	1
597	**FT Sensor Offset Setting [1]** 1: Offset Setting	-	0

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6.2.2 Input Register (Read Only)

Address DEC	Description.	Scale
0	**Product ID**	-
1	**Firmware version**	-
2	**Whether the gripper joint moves** Holding Register 369 ref • 0: Stop • 1: Movement	-
3	**Arrive Joint Target Value** Verify Target Value Arrival for All Joints See Holding Register 339 to 358 • 0: Not reached • 1: Arrival	-
4	**Currently Active Blend Number** • Indicates the Execution Order of the Currently Active Blend	-
5	**Blend Status** • Displaying Status Based on Blend Behavior • 0: Stop • 1: In Progress • 2: Done	-
6	**Motor 1 Current Position Value** • Unit: 0.1°	x10
7	**Motor 2 Current Position Value**	x10
8	**Motor 3 Current Position Value**	x10
9	**Motor 4 Current Position Value**	x10
10	**Motor 5 Current Position Value**	x10
11	**Motor 6 Current Position Value**	x10
12	**Motor 7 Current Position Value**	x10
13	**Motor 8 Current Position Value**	x10
14	**Motor 9 Current Position Value**	x10
15	**Motor 10 Current Position Value**	x10

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16	**Motor 11 Current Position Value**	x10
17	**Motor 12 Current Position Value**	x10
18	**Motor 13 Current Position Value**	x10
19	**Motor 14 Current Position Value**	x10
20	**Motor 15 Current Position Value**	x10
21	**Motor 16 Current Position Value**	x10
22	**Motor 17 Current Position Value**	x10
23	**Motor 18 Current Position Value**	x10
24	**Motor 19 Current Position Value**	x10
25	**Motor 20 Current Position Value**	x10
26	**Motor 1 Current Value** • Unit: 1 mA	-
27	**Motor 2 Current Value**	-
28	**Motor 3 Current Value**	-
29	**Motor 4 Current Value**	-
30	**Motor 5 Current Value**	-
31	**Motor 6 Current Value**	-
32	**Motor 7 Current Value**	-
33	**Motor 8 Current Value**	-
34	**Motor 9 Current Value**	-
35	**Motor 10 Current Value**	-
36	**Motor 11 Current Value**	-
37	**Motor 12 Current Value**	-
38	**Motor 13 Current Value**	-
39	**Motor 14 Current Value**	-
40	**Motor 15 Current Value**	-
41	**Motor 16 Current Value**	-
42	**Motor 17 Current Value**	-
43	**Motor 18 Current Value**	-
44	**Motor 19 Current Value**	-
45	**Motor 20 Current Value**	-
46	**Motor 1 Current Speed Value** • Unit: 1 RPM	-
47	**Motor 2 Current Speed Value**	-

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48	**Motor 3 Current Speed Value**	-
49	**Motor 4 Current Speed Value**	-
50	**Motor 5 Current Speed Value**	-
51	**Motor 6 Current Speed Value**	-
52	**Motor 7 Current Speed Value**	-
53	**Motor 8 Current Speed Value**	-
54	**Motor 9 Current Speed Value**	-
55	**Motor 10 Current Speed Value**	-
56	**Motor 11 Current Speed Value**	-
57	**Motor 12 Current Speed Value**	-
58	**Motor 13 Current Speed Value**	-
59	**Motor 14 Current Speed Value**	-
60	**Motor 15 Current Speed Value**	-
61	**Motor 16 Current Speed Value**	-
62	**Motor 17 Current Speed Value**	-
63	**Motor 18 Current Speed Value**	-
64	**Motor 19 Current Speed Value**	-
65	**Motor 20 Current Speed Value**	-
66	**Motor 1 Current Temperature Value** • Unit: 0.1 C°	x10
67	**Motor 2 Current Temperature Value**	x10
68	**Motor 3 Current Temperature Value**	x10
69	**Motor 4 Current Temperature Value**	x10
70	**Motor 5 Current Temperature Value**	x10
71	**Motor 6 Current Temperature Value**	x10
72	**Motor 7 Current Temperature Value**	x10
73	**Motor 8 Current Temperature Value**	x10
74	**Motor 9 Current Temperature Value**	x10
75	**Motor 10 Current Temperature Value**	x10
76	**Motor 11 Current Temperature Value**	x10
77	**Motor 12 Current Temperature Value**	x10
78	**Motor 13 Current Temperature Value**	x10
79	**Motor 14 Current Temperature Value**	x10

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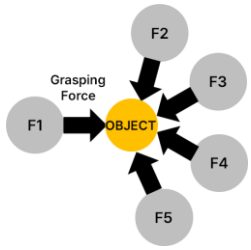
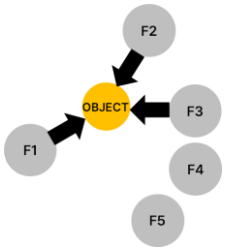
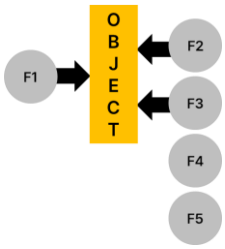
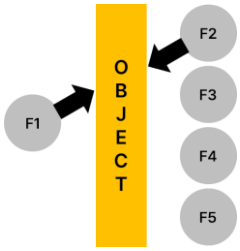
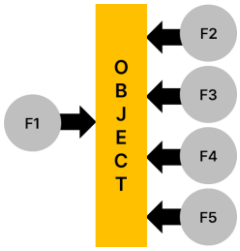
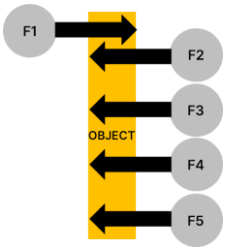
80	**Motor 15 Current Temperature Value**	x10
81	**Motor 16 Current Temperature Value**	x10
82	**Motor 17 Current Temperature Value**	x10
83	**Motor 18 Current Temperature Value**	x10
84	**Motor 19 Current Temperature Value**	x10
85	**Motor 20 Current Temperature Value**	x10
127	**GPIO Input1 Status** • 0: 0 V • 1: 24 V	-
128	**Fingertip F/T Sensor 1 Fx** • Unit: 0.1 N	x10
129	**Fingertip F/T Sensor 1 Fy** • Unit: 0.1 N	x10
130	**Fingertip F/T Sensor 1 Fz** • Unit: 0.1 N	x10
131	**Fingertip F/T Sensor 1 Tx** • Unit: 1 mNm	
132	**Fingertip F/T Sensor 1 Ty** • Unit: 1 mNm	
133	**Fingertip F/T Sensor 1 Tz** • Unit: 1 mNm	
134	**Fingertip F/T Sensor 2 Fx**	x10
135	**Fingertip F/T Sensor 2 Fy**	x10
136	**Fingertip F/T Sensor 2 Fz**	x10
137	**Fingertip F/T Sensor 2 Tx**	
138	**Fingertip F/T Sensor 2 Ty**	
139	**Fingertip F/T Sensor 2 Tz**	
140	**Fingertip F/T Sensor 3 Fx**	x10
141	**Fingertip F/T Sensor 3 Fy**	x10
142	**Fingertip F/T Sensor 3 Fz**	x10
143	**Fingertip F/T Sensor 3 Tx**	
144	**Fingertip F/T Sensor 3 Ty**	
145	**Fingertip F/T Sensor 3 Tz**	
146	**Fingertip F/T Sensor 4 Fx**	x10
147	**Fingertip F/T Sensor 4 Fy**	x10

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148	**Fingertip F/T Sensor 4 Fz**	x10
149	**Fingertip F/T Sensor 4 Tx**	
150	**Fingertip F/T Sensor 4 Ty**	
151	**Fingertip F/T Sensor 4 Tz**	
152	**Fingertip F/T Sensor 5 Fx**	x10
153	**Fingertip F/T Sensor 5 Fy**	x10
154	**Fingertip F/T Sensor 5 Fz**	x10
155	**Fingertip F/T Sensor 5 Tx**	
156	**Fingertip F/T Sensor 5 Ty**	
157	**Fingertip F/T Sensor 5 Tz**	

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6.2.3 Grasp Mode

21		A grasping algorithm that follows the center point where the five fingers cross, based on force
22		A grasping algorithm that follows the center point where three fingers cross, based on force (Finger 1 and Finger 2 and Finger 3)
23		A grasping algorithm that follows the force-based tracking, where one finger and its corresponding two fingers are driven in parallel (Finger 1 and Finger2 and Finger3)
24 25 26 27		A grasping algorithm that follows the center point where two fingers cross, based on force (Finger 1 and Finger2 - Finger5)
28		A grasping algorithm that follows the force-based tracking, where one finger and its corresponding four fingers are driven in parallel.
29		A grasping algorithm where all five fingers operate to wrap around the target object based on force

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